

Elevator Simulation - Scheduler Comparison

Nearest Car vs Round Robin vs Zone Based across all scenarios

baseline - Standard case - 2 elevators, staggered arrivals

Elevators: 2 Floors: 60 Capacity/elevator: 10

Metric	Nearest Car	Round Robin	Zone Based
Completed	3/3	3/3	3/3
Avg wait time	6 ticks	2.7 ticks	2.7 ticks
Max wait time	18 ticks	8 ticks	8 ticks
Avg total time	40.3 ticks	57.7 ticks	57.7 ticks
Max total time	49 ticks	89 ticks	89 ticks
Utilization	E0: 50%, E1: 38%	E0: 99%, E1: 36%	E0: 99%

scenario_single_elevator - Edge case - only 1 elevator serving 6 passengers

Elevators: 1 Floors: 60 Capacity/elevator: 5

Metric	Nearest Car	Round Robin	Zone Based
Completed	3/6	3/6	3/6
Avg wait time	1 ticks	1 ticks	1 ticks
Max wait time	3 ticks	3 ticks	3 ticks
Avg total time	36.3 ticks	36.3 ticks	36.3 ticks
Max total time	53 ticks	53 ticks	53 ticks
Utilization	E0: 100%	E0: 100%	E0: 100%

scenario_many_elevators - Edge case - 8 elevators for only 3 passengers (under-utilised)

Elevators: 8 Floors: 60 Capacity/elevator: 10

Metric	Nearest Car	Round Robin	Zone Based
Completed	2/3	3/3	3/3
Avg wait time	6.5 ticks	13.7 ticks	13.7 ticks
Max wait time	13 ticks	28 ticks	28 ticks
Avg total time	50.5 ticks	52.7 ticks	52.7 ticks
Max total time	58 ticks	58 ticks	58 ticks
Utilization	E0: 100%	E0: 58%, E1: 57%, E2: 44%	E0: 58%, E1: 44%, E3: 57%

scenario_capacity_overflow - Edge case - 8 passengers arrive at t=0, elevators hold only 2 each

Elevators: 2 Floors: 60 Capacity/elevator: 2

Metric	Nearest Car	Round Robin	Zone Based
Completed	5/8	4/8	2/8
Avg wait time	11.2 ticks	0 ticks	0 ticks
Max wait time	56 ticks	0 ticks	0 ticks
Avg total time	46.2 ticks	33 ticks	43 ticks

Max total time	99 ticks	48 ticks	48 ticks
Utilization	E0: 100%, E1: 100%	E0: 100%, E1: 100%	E0: 100%

scenario_simultaneous_same_floor - Edge case - 6 passengers all on floor 1 going to different floors

Elevators: 2 Floors: 20 Capacity/elevator: 5

Metric	Nearest Car	Round Robin	Zone Based
Completed	6/6	6/6	6/6
Avg wait time	0 ticks	0 ticks	6.2 ticks
Max wait time	0 ticks	0 ticks	37 ticks
Avg total time	9.7 ticks	9.7 ticks	16 ticks
Max total time	18 ticks	18 ticks	48 ticks
Utilization	E0: 19%, E1: 11%	E0: 14%, E1: 19%	E0: 49%

scenario_bidirectional - Edge case - simultaneous up and down traffic across 3 elevators

Elevators: 3 Floors: 30 Capacity/elevator: 4

Metric	Nearest Car	Round Robin	Zone Based
Completed	8/8	8/8	8/8
Avg wait time	10.4 ticks	10.4 ticks	10.4 ticks
Max wait time	28 ticks	28 ticks	28 ticks
Avg total time	37.4 ticks	34.6 ticks	33.4 ticks
Max total time	57 ticks	57 ticks	57 ticks
Utilization	E0: 57%, E1: 51%	E0: 57%, E1: 48%, E2: 29%	E0: 29%, E1: 36%, E2: 57%

scenario_express_elevator - Bonus - 1 express elevator serving floors {1,20,40,60} only

Elevators: 3 Floors: 60 Capacity/elevator: 5

Metric	Nearest Car	Round Robin	Zone Based
Completed	5/7	6/7	4/7
Avg wait time	4.8 ticks	15.2 ticks	14 ticks
Max wait time	18 ticks	42 ticks	38 ticks
Avg total time	37 ticks	46.7 ticks	49 ticks
Max total time	58 ticks	67 ticks	77 ticks
Utilization	E0: 44%, E1: 34%, E2: 100%	E0: 100%, E1: 66%, E2: 77%	E0: 100%, E1: 77%

scenario_morning_rush - Morning rush - 15 passengers all from floor 1 heading up, overflow at t=0

Elevators: 3 Floors: 30 Capacity/elevator: 4

Metric	Nearest Car	Round Robin	Zone Based
Completed	15/15	15/15	12/15
Avg wait time	3 ticks	8.6 ticks	20.7 ticks
Max wait time	15 ticks	47 ticks	47 ticks

Avg total time	18.1 ticks	23.7 ticks	33.3 ticks
Max total time	44 ticks	76 ticks	71 ticks
Utilization	E0: 45%, E1: 16%, E2: 24%	E0: 65%, E1: 71%, E2: 76%	E0: 100%

scenario_lunchtime - Lunchtime - 15 passengers on upper floors all heading down to floor 1, overflow

Elevators: 3 Floors: 30 Capacity/elevator: 4

Metric	Nearest Car	Round Robin	Zone Based
Completed	15/15	15/15	13/15
Avg wait time	18.1 ticks	22.9 ticks	15.5 ticks
Max wait time	44 ticks	68 ticks	49 ticks
Avg total time	39.4 ticks	53.8 ticks	35 ticks
Max total time	73 ticks	97 ticks	67 ticks
Utilization	E0: 73%, E1: 32%, E2: 48%	E0: 97%, E1: 44%, E2: 48%	E0: 16%, E1: 67%, E2: 100%

scenario_townhall - Townhall - 15 passengers converging on floor 15, mixed up/down traffic, overflow

Elevators: 3 Floors: 30 Capacity/elevator: 4

Metric	Nearest Car	Round Robin	Zone Based
Completed	15/15	15/15	15/15
Avg wait time	14.1 ticks	14.1 ticks	18.1 ticks
Max wait time	28 ticks	28 ticks	52 ticks
Avg total time	24.6 ticks	27.8 ticks	28.5 ticks
Max total time	43 ticks	43 ticks	67 ticks
Utilization	E0: 44%, E1: 18%, E2: 34%	E0: 38%, E1: 42%, E2: 44%	E0: 26%, E1: 22%, E2: 67%

scenario_ceo_visit - CEO visit - morning rush with express lane reserved for executives (floors {1,25

Elevators: 3 Floors: 30 Capacity/elevator: 4

Metric	Nearest Car	Round Robin	Zone Based
Completed	16/16	16/16	12/16
Avg wait time	3.8 ticks	23.9 ticks	43.3 ticks
Max wait time	15 ticks	57 ticks	73 ticks
Avg total time	19.6 ticks	39.9 ticks	57.9 ticks
Max total time	41 ticks	79 ticks	89 ticks
Utilization	E0: 42%, E1: 16%, E2: 29%	E0: 79%, E1: 77%, E2: 24%	E0: 100%